

out lets of category 13A in the same room. be connected to the same cct breaker in the same DB.

However, this is not the case in non-domestic premises. All socket outlets shall not be less than 2M away from one another horizontally

Busbar: A busbar in domestic distribution refers to thick strips of copper / Aluminium that conducts electricity within a switch board, DB or other electrical apparatus. It can also be found in substations inside feeder pillars. Each busbar has a current carrying capacity that is determined by its size and the type of conducting material employed. hollow tubes, Busbars are typically flat strips or those shapes are adopted because they allow heat to dissipate more efficiently due to the high surface area to cross-sectional area ratio.

DB and Consumer Unit:

A DB is a mounting enclosure for multiple electrical cct breakers, these are generally placed in columns and aligned ^{with} small single & boxes with cct breaker in just one row. (and three rows for 3-Ø). DB are ~~usually~~ ^{typically} found in central locations ~~within~~ ^{within} buildings as much as possible and often serve as the point at which electricity is distributed within a building. Cct breakers can be used to manually de-energise electrical ccts when the downstream wiring is being serviced.

PVC Insulated and Sheathed Cables. This type of cable is used in commercial and domestic installations, which may be clipped direct to a surface, sunk in plaster or installed in conduit or trunking. It is the simplest and the ~~last~~ expensive. The conductors are covered with a colour-coded PVC insulator and then contained singly or with others in PVC outer sheath.

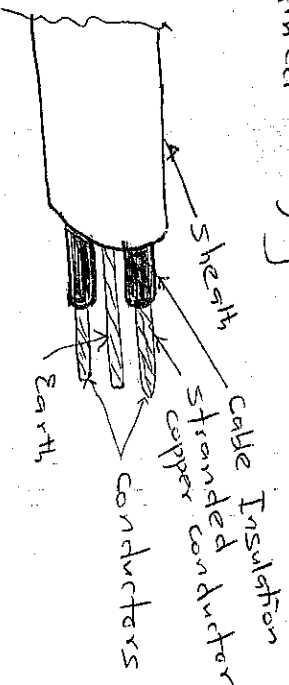


Fig. 7.0(a): A twin and earth PVC insulated and sheathed cable

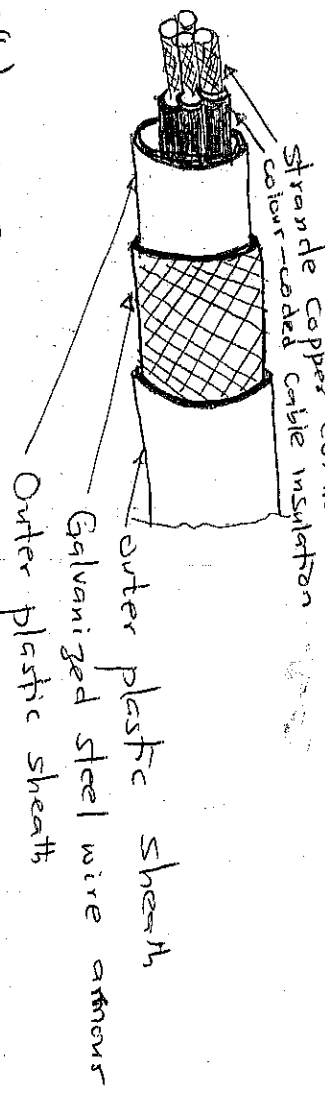


Fig. 7.0(b) A four-core PVC/SWA cable

PVC/SWA Cable: PVC insulated steel wire armoured cable are used for wiring underground cables b/w buildings, for mains supplies to dwellings, rising sub-mains and industrial installations. They are used where some mechanical protection of the cable is required. The galvanized steel wire armour provides the mechanical protection and the outer ^{plastic} sheath protects them from corrosion.

Mineral Insulated (MI) Cable: This type of cable has a seamless copper sheath which makes it water proof and fire and corrosion-resistant. This Xtrac makes it the only cable choice for ~~her~~ gardens or high-temp installations such as oil refineries and mechanical

Page 23
Protection for the cables. Conduit wiring is very popular in ~~both~~ commercial, domestic and industrial applications. There are three ^{basic} types of conduit used in electrical installations which include steel, PVC and flexible.

Cable Parameters

GO TO 23(5)

To be discussed in class

- (i) Length of run
 - (ii) Ambient temp
 - (iii) Grouping of ccts.
 - (iv) Disposition of cables
 - (v) Type of sheathing
 - (vi) Rating
 - (vii) Voltage drop
 - (viii) Class of excess current protection
- $$\begin{aligned} & \text{supply} < \begin{matrix} 2.5\% \\ V = V \\ 6\% \end{matrix} \\ & \text{distribution} < \begin{matrix} 3\% \text{ lighting} \\ 5\% \text{ power} \end{matrix} \\ & \text{voltage drop} \\ & V = \text{voltage variation} \end{aligned}$$
- 2.5% if nominal (same) of fluctuation/variation

Cable Selection Parameters

Parameters for selection and utilization of cable system for wiring a building differs from that of design and construction of cable. In selection and utilization of cables, certain relevant factors need to be considered, probably the most significant factor in cable selection is eight factors listed above.

and installation are the following:
All cables have certain resistance which give rise to heat when current passes through them, and the size, and the material, ^{of conductor} load connected and temp determines its withstand ^{current carrying capacity} of the cable.

BS6004 PVC insulated cables are suitable for operation as long as the temp of the conductor doesn't exceed 70°C.

Single pre insulated conductors are usually drawn in to the installed conduit to complete the installation. Having decided upon the type, size and number of cables required for a final circuit, it is then necessary to select the appropriate size of conduit to accommodate those cables.

The tables in Appendix 5 of the On Site Guide describe a "factor system" for determining the size of ~~cable~~ ^{conduit} to required to enclose number of conductors as seen in table 1(a) and 1(b) given below. The method is as follows

- i. Identify the cable factor for the particular size of conductor from table 1(a).
- ii. Multiply the cable factor by the number of conductor (with the same size) to give sum of cable factors
- iii. Identify the appropriate part of the conductor factor given by the length of run, and number of bends (table 1b)
- iv. The correct size of ~~conduit~~ ^{conduit} to accommodate the cable is that is the conduit size which has size that is equal to or greater than the ^{sum of} cable factor, but ^{the closest} size should be, ~~preferably~~ ^{preferably} be selected to minimise space as well as cost.

Example

Four 1.5mm² PVC insulated cables are to be drawn into a plastic conduit which is 5m long between boxes and contains two bends. A 4mm² PVC insulated CPC is also included. Determine the ^{optimum} appropriate size of conduit to contain these conductors.

and is ideal for switch drops or for housing auxiliary cts such as telephone or audio equipment wiring

A Skirting trunking: A trunking manufactured from PVC or steel and in ^{it} the shape of a skirting board is frequently used in commercial buildings such as hospitals, labs and offices. The trunking is fitted around the walls of a room and contains a wiring for socket outlets and telephone pts which are mounted on the lids.